

USE CASE SPECIFICATION DOCUMENT

Library Management System

Project Name: Library Management System

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Date: June 2026

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5. Summary

1. Introduction

The Library Management System is a comprehensive web-based application engineered to automate and optimize standard library operations. This system streamlines book inventory management, member tracking, book issuance and returns, automated fine assessment, and executive report generation. By providing a structured architecture for data management, the application reduces manual overhead and eliminates operational errors.

This document provides detailed Use Case Specifications detailing functional requirements, user workflows, preconditions, postconditions, and conversational branches across multiple system actors. It acts as the definitive functional reference for engineering, testing, and deployment phases.

2. Actors

2.1 Administrator

The Administrator oversees systemic security, configuration management, and baseline personnel structures. They possess overarching analytical view access across all functional business verticals.

Key Responsibilities:

- Manage global user entities and roles.
- Provision and audit Librarian profiles.
- Review aggregated system operational reports and analytics.
- Configure baseline structural and functional application parameters.

2.2 Librarian

The Librarian is the standard executive agent executing day-to-day transactional business logic inside the library ecosystem.

Key Responsibilities:

- Maintain the catalog system (Add, Update, Delete Books).
- Register and update Member records.
- Coordinate core transactions (Issue Book, Return Book).
- Trigger manual or periodic report distributions.

2.3 Student / Member

The Student/Member represents the end consumer consuming the library's physical inventory resources via the digital self-service dashboard.

Key Responsibilities:

- Authenticate securely through individual account credentials.
- Query inventory catalog using advanced filtering options.
- Monitor current active borrows and historical interactions.
- Review outstanding financial fines or disciplinary penalties.
- Modify self-managed personal identification profiles.

3. Use Case Diagram Description

The system boundaries delineate three distinct macro-actors interacting with individual operational endpoints. Below is the functional mapping summarizing actor-to-feature execution pathways:

- **Administrator:** Login, Manage Users, Manage Librarians, Generate Reports, View Analytics Dashboard.
- **Librarian:** Login, Add Book, Update Book, Delete Book, Register Member, Issue Book, Return Book, Calculate Fine, Generate Reports.
- **Student / Member:** Login, Search Books, View Profile, View Borrowing History, View Fine Information.

4. Use Case Specifications

UC-01: User Login	
Primary Actor	Administrator, Librarian, Student
Goal	Access the application securely.
Preconditions	The user account exists and is properly registered in the database.
Main Flow	1. User opens the system login page. 2. User enters their registered email address. 3. User enters their password. 4. User clicks the "Login" button. 5. System validates credentials against stored security records. 6. System redirects the authenticated user to their role-specific dashboard.
Alternative Flow	A1: Invalid credentials entered 1. System flags mismatched username/password combination. 2. System halts entry and displays an explicit authentication error message. 3. User is prompted to clear or re-enter valid data fields.
Postconditions	User successfully establishes an active secure session in the system.

UC-02: Add New Book

Primary Actor	Librarian
Goal	Add a new unique book record to the library inventory.
Preconditions	Librarian is securely logged in.
Main Flow	1. Librarian navigates to and selects "Add Book" from the navigation catalog. 2. System displays the book asset metadata form fields. 3. Librarian enters: Title, Author, ISBN, Category, and Inventory Quantity. 4. Librarian submits the form. 5. System validates required data fields and structural formats. 6. System commits and stores book information into the persistent database.
Postconditions	The new book record is successfully created and inventory counts are incremented.

UC-03: Update Book

Primary Actor	Librarian
Goal	Modify attribute datasets of existing book listings.
Preconditions	The requested book record exists; Librarian is logged in.
Main Flow	1. Librarian searches for the intended book entity. 2. Librarian opens the metadata edit panel interface. 3. Librarian overwrites fields with updated text data strings. 4. Librarian clicks "Save Changes". 5. System updates database records synchronously.
Postconditions	Book database row attributes reflect modified inputs successfully.

UC-04: Delete Book

Primary Actor	Librarian
Goal	Remove a physical or digital book row entity completely from inventory tracking.
Preconditions	Target book exists; Librarian is logged in; No current active transactional borrowing records exist for this asset.
Main Flow	1. Librarian searches for the specific target book asset. 2. Librarian clicks the system "Delete" item interface trigger. 3. Librarian confirms intent via the hard alert modal check pop-up. 4. System deletes the database registry entry.
Postconditions	The system clears the book record successfully.

UC-05: Search Book

Primary Actor	Student, Librarian
Goal	Query catalog datasets efficiently.
Preconditions	User session is established and active.
Main Flow	1. User opens the central inventory search interface. 2. User provides search strings inside: Title, Author, ISBN, or Category filters. 3. User triggers the search query button execution. 4. System executes query filters and parses indices. 5. System generates and renders matching entity row collections.
Postconditions	Dynamic query list returns context results corresponding to input filters.

UC-06: Register Member

Primary Actor	Librarian
Goal	Create a new student or general member account profiling target.
Preconditions	Librarian session is established and active.
Main Flow	1. Librarian initiates member registration flow interface panel. 2. Librarian keys in Member profile datasets (Name, Contact Email, Assigned Role Group). 3. Librarian submits user structural form dataset. 4. System parses schemas and asserts unique data field requirements. 5. System records entry row and instantiates default password profiles.
Postconditions	New user profile is populated into member tracking tables successfully.

UC-07: Issue Book

Primary Actor	Librarian
Goal	Create a validated ledger borrow entry tracking an asset loan.
Preconditions	Target member profile exists; Book is actively available in stock; Librarian is logged in.
Main Flow	1. Librarian searches and selects the requesting Member profile. 2. Librarian searches and selects the physical target book entity. 3. System checks actual real-time physical warehouse availability. 4. Librarian clicks "Issue Book". 5. Librarian keys in Issue Date and structural Return Due Date parameters. 6. Librarian commits transactional log entry row to system database. 7. System automatically reduces actual asset available unit inventory count.
Alternative Flow	A1: Book unavailable in stock 1. System rejects issue lifecycle request. 2. System raises warning indicating target asset allocation count stands at 0. 3. Transaction processes abort.
Postconditions	A new transaction borrow entry row links the book lifecycle state to the member.

UC-08: Return Book

Primary Actor	Librarian
Goal	Process a returned asset item and update historical ledgers.
Preconditions	An active borrow transaction record maps to this asset.
Main Flow	1. Librarian locates target active ledger borrow entity row. 2. Librarian clicks "Return Book" operation trigger. 3. Librarian validates and enters absolute Return Date values. 4. System triggers fine engine rules calculations automatically. 5. System increments available stock quantities. 6. Transaction records close out and flag state adjustments to complete.
Postconditions	The asset object records back to base inventory; financial parameters resolve.

UC-09: Fine Calculation

Primary Actor	System
Goal	Automatically compute dynamic fines based on late returns.
Preconditions	Actual recorded check-in date exceeds initialized loan Due Date value.
Main Flow	1. System computes raw time delta integer representing overdue day lengths. 2. System pulls standardized variable daily fine baseline parameters. 3. System processes programmatic functional data mathematical equation calculations. 4. System generates a permanent static record binding fee rows to individual Member accounts.
Mathematical Formula	$\text{Fine} = \text{Overdue Days} \times \text{Daily Fine Rate}$
Postconditions	Financial fine penalty values are computed and appended to target database rows.

UC-10: View Borrowing History

Primary Actor	Student
Goal	Access individual chronological history logs tracking asset checkouts.
Preconditions	Student account authentication session is active.
Main Flow	1. Student accesses personal user profile overview panel. 2. Student selects the dedicated "Borrowing History" data view action. 3. System queries transactional databases filter-matched to active Member ID. 4. System populates historic tables matching tracking parameters.
Postconditions	Personal history logs render to the interface frame successfully.

UC-11: View Fine Information

Primary Actor	Student
Goal	Track systemic accounting statements of accumulated fees due.
Preconditions	Student account authentication session is active.
Main Flow	1. Student logs into internal landing interface window. 2. Student clicks into dedicated "Fines Section" node interface. 3. System pulls tracking account entities corresponding to data indexes. 4. Student reviews line-item lists identifying open ledger charges.
Postconditions	The student gains direct access transparency viewing all open personal liabilities.

UC-12: Generate Reports

Primary Actor	Administrator, Librarian
Goal	Extract operational metrics datasets into structured reporting formats.
Preconditions	Authorized personnel level authentication session is active.
Main Flow	1. User accesses system Report Module panels. 2. User designates specialized structural format outputs (Daily, Weekly, Monthly, Fine, Borrowing Report). 3. User executes the "Generate" report command button. 4. System runs server compilation aggregations compiling layout profiles. 5. System yields options to execute static Print commands or compile local file Downloads.
Postconditions	A data-driven snapshot file exports from the transactional system environment.

UC-13: Manage Users

Primary Actor	Administrator
Goal	Administer system actor profile attributes and infrastructure privileges.
Preconditions	Administrator level access security validation token is verified active.
Main Flow	1. Administrator calls user management dashboard. 2. System lists registered credential entity types across roles. 3. Administrator commands modification functions (Add, Edit, Delete Users). 4. Administrator confirms changes and hits save controls.
Postconditions	Underlying systemic credential configurations refresh tracking data accurately.

UC-14: View Analytics Dashboard

Primary Actor	Administrator
Goal	Monitor core Key Performance Indicators (KPIs) through high-level visualization panels.
Preconditions	Administrator session validation token is verified active.
Main Flow	1. Administrator enters system application dashboard index. 2. System initializes metrics and calculates database counts, populating fields: ◦ Total Books Inventory Count ◦ Total Registered Members Base ◦ Active Outstanding Borrowings ◦ Gross Accumulated Fine Collections ◦ Monthly Activity Velocity Statistics 3. Administrator reviews systemic analytical values.
Postconditions	Live operational parameters display correctly on the workspace dashboard.

5. Summary

The Use Case Specification document details the underlying functional blueprint regulating the Library Management System application architecture. By grouping operational scopes across distinct actors (Administrators, Librarians, and Students), the application secures access control patterns while minimizing structural manual workflow friction. These behavioral use cases represent standard blueprints used across subsequent architectural system development lifecycles, establishing core compliance frameworks for developers, database designers, quality assurance engineers, and administrative project stakeholders.